Supplementary Materials

Adaptive Pair Selection and Score Estimation in LLM-Based Comparative Judgment of Writing

Supplementary Appendix A

Analytic Writing Traits and Guidance

The following trait-specific guidance was provided to the LLM during comparative judgment. The wording was based on the ELLIPSE analytic writing rubrics (Crossley et al., 2023).

Cohesion

Construct definition. Cohesion concerns text organization and the use of cohesive devices to connect ideas within and across sentences and paragraphs.

Scoring considerations. Evaluate overall organizational control and logical sequencing; the range, appropriateness, and effectiveness of cohesive devices such as reference and transitional words or phrases; connections within and across sentences and paragraphs; continuity and appropriate overlap of ideas; and whether cohesive links become repetitive, mechanical, inaccurate, or unclear.

Construct boundaries. Focus on organization, sequencing, cohesive devices, and continuity across ideas. Treat sentence formation and word order as primarily part of Syntax, individual-word precision as primarily part of Vocabulary, and spelling, capitalization, and punctuation as part of Conventions. Do not score idea quality, originality, persuasiveness, essay length, or paragraph count as evidence of Cohesion. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Syntax

Construct definition. Syntax concerns the range and control of sentence structures.

Scoring considerations. Evaluate variation and flexibility across simple, compound, and complex sentences; sentence formation and word order; appropriateness of syntactic choices; and the frequency and seriousness of structural errors, particularly as sentence complexity increases.

Construct boundaries. Focus on sentence structures, sentence formation, and word order. Treat lexical range and precision as primarily part of Vocabulary, morphology and grammatical usage as primarily part of Grammar, spelling, capitalization, and punctuation as part of Conventions, and organization and connections across sentences and paragraphs as primarily part of Cohesion. Do not reward lexical sophistication, content quality, or complexity of ideas merely because they occur in complex sentences. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Vocabulary

Construct definition. Vocabulary concerns the precision, range, and appropriateness of word choice.

Scoring considerations. Evaluate breadth and flexibility of vocabulary; precision of word choice; appropriate use of topic-related terms and less-common lexical items; control of word choice and word forms where the official rubric treats them as lexical evidence; and whether lexical inaccuracies make meaning imprecise or distort it.

Construct boundaries. Focus on individual word choice, lexical range, precision, and appropriateness. Treat multiword expressions as primarily part of Phraseology, sentence structure as primarily part of Syntax, and spelling, capitalization, and punctuation as part of Conventions. Do not reward uncommon words merely for being uncommon, do not treat topic-related vocabulary as a separate task-fulfillment score, and do not automatically reassign every word-form error to Grammar when the rubric treats word forms as lexical evidence. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Phraseology

Construct definition. Phraseology concerns the use and control of multiword expressions, especially idioms, collocations, lexical bundles, and other recurring phrasal patterns.

Scoring considerations. Evaluate the variety and flexibility of phrases; appropriateness and accuracy of multiword combinations; the ability to use phrases to convey precise or subtle meanings; noticeable repetition or misuse of phrases; and overreliance on memorized chunks or simple phrasal patterns.

Construct boundaries. Focus on multiword expressions and recurring phrasal patterns. Treat individual-word range and precision as primarily part of Vocabulary, sentence structure as primarily part of Syntax, and morphology and grammatical usage as primarily part of Grammar. Do not reward an idiom, collocation, or unusual phrase merely for being conspicuous; it should be appropriate and accurately used. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Grammar

Construct definition. Grammar concerns control of morphology and grammatical usage.

Scoring considerations. Evaluate the frequency and pervasiveness of errors in grammar and usage and the overall degree of grammatical control demonstrated across the essay. The official rubric defines this trait primarily through the amount of grammatical and usage error, so do not introduce additional subcriteria that are not present in the source rubric.

Construct boundaries. Focus on morphology and grammatical usage. Treat sentence-structure range, sentence formation, and word order as primarily part of Syntax, spelling, capitalization, and punctuation as part of Conventions, and lexical range and word choice as primarily part of Vocabulary. Do not use sophistication in sentence structure, vocabulary, phrase choice, or content as evidence of grammatical control by itself. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Conventions

Construct definition. Conventions concerns control of spelling, punctuation, and capitalization.

Scoring considerations. Evaluate the consistency and appropriateness of spelling, punctuation, and capitalization; the frequency of errors; and whether those errors are negligible, noticeable, distracting, frequent, or pervasive across the essay.

Construct boundaries. Focus on spelling, punctuation, and capitalization. Treat morphology and grammatical usage as primarily part of Grammar, sentence-structure range and formation as primarily part of Syntax, and organization and connections across sentences and paragraphs as primarily part of Cohesion. Do not use punctuation as evidence of syntactic range or complexity; judge it here as a written convention. These boundaries are intended to reduce cross-trait halo effects, not to imply that language features never overlap.

Supplementary Appendix B

Comparative-Judgment Instructions

The following template was used to generate the LLM-based comparative judgments. The {TRAIT_DESCRIPTION} placeholder was populated with the corresponding construct definition, scoring considerations, and construct boundaries from Supplementary Appendix A.

```
You are an expert writing assessor. Decide which writing sample demonstrates
stronger performance on the specified writing trait.

TRAIT: {TRAIT_NAME}

TRAIT GUIDANCE:
{TRAIT_DESCRIPTION}

ASSESSMENT RULES:
- Evaluate SAMPLE_1 and SAMPLE_2 on {TRAIT_NAME} before comparing them.
- Consider the overall pattern of evidence relevant to the trait, including both
strengths and limitations.
- Judge only {TRAIT_NAME}. Do not compensate for strengths or weaknesses that
primarily belong to another writing trait.
- Compare the samples directly. Do not assign or infer rubric scores.
- Do not prefer a sample merely because it is longer, takes a particular
position, uses more sophisticated subject matter, or is more interesting.
- Use topic or content information only when it is directly relevant to
interpreting evidence for {TRAIT_NAME}.
- A decision is required; do not return a tie.
- Do not default to the sample presented first.

=== SAMPLE_1 ===
{SAMPLE_1}
=== END SAMPLE_1 ===

=== SAMPLE_2 ===
{SAMPLE_2}
=== END SAMPLE_2 ===

FINAL DECISION:
Return only one of these XML tags, with no explanation or other text:

<BETTER_SAMPLE>SAMPLE_1</BETTER_SAMPLE>
OR
<BETTER_SAMPLE>SAMPLE_2</BETTER_SAMPLE>
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Supplementary Appendix C

TrueSkill Updates and Hybrid Pair Selection

The adaptive pair selection strategies used TrueSkill, a Bayesian rating system designed for efficient sequential updating after each observed comparison (Herbrich et al., 2007). TrueSkill was used only to select pairs; the reported CJ scores were estimated separately. At each point in the replay, essay i was represented by a normal distribution with mean μ_i and standard deviation σ_i. The mean μ_i represented the essay's current location on the quality scale, and σ_i represented uncertainty about that location. All essays received the standard TrueSkill prior, with $\mu_i = 25$ and $\sigma_i = 25/3$, and the performance variability parameter was fixed at the conventional value $\beta_{\mathrm{TS}} = 25/6$ (Herbrich et al., 2007). The parameter β_{TS} represents the standard deviation of comparison-specific performance variation. Independent variation in both performances therefore contributes $2\beta_{\mathrm{TS}}^2$ to the variance of their difference. For essays i and j, the preference probability used for pair selection was

$$p_{ij}^{\mathrm{TS}} = \Phi\left(\frac{\mu_i - \mu_j}{\sqrt{\sigma_i^2 + \sigma_j^2 + 2\beta_{\mathrm{TS}}^2}}\right), \tag{C1}$$

where Φ is the standard normal cumulative distribution function. Probabilities near .50 therefore indicated that the current TrueSkill information did not strongly favor either essay, whereas probabilities farther from .50 indicated a more clearly separated pair. The corresponding ambiguity utility was

$$U_{0,ij} = p_{ij}^{\mathrm{TS}}\left(1 - p_{ij}^{\mathrm{TS}}\right), \tag{C2}$$

which reaches its maximum of .25 when $p_{ij}^{\mathrm{TS}} = .50$.

After each revealed judgment, TrueSkill updated only the two essays in that comparison. For a comparison in which essay i was preferred to essay j, define

$$c = \sqrt{2\beta_{\mathrm{TS}}^2 + \sigma_i^2 + \sigma_j^2}, \quad t = \frac{\mu_i - \mu_j}{c}, \tag{C3}$$

and

$$V(t) = \frac{\phi(t)}{\Phi(t)}, \quad W(t) = V(t)\{V(t) + t\}, \tag{C4}$$

where ϕ is the standard-normal density. For the binary, no-draw comparisons used here, the sequential update was

$$\begin{aligned}
\mu_i' &= \mu_i + \frac{\sigma_i^2}{c} V(t), \\
\mu_j' &= \mu_j - \frac{\sigma_j^2}{c} V(t), \\
(\sigma_i')^2 &= \sigma_i^2 \left\{1 - \frac{\sigma_i^2}{c^2} W(t)\right\}, \\
(\sigma_j')^2 &= \sigma_j^2 \left\{1 - \frac{\sigma_j^2}{c^2} W(t)\right\}.
\end{aligned} \tag{C5}$$

Thus, each judgment increased the estimated location of the preferred essay, decreased that of the other essay, and updated the uncertainty associated with both.

For the random, p50, and Pollitt (p33/67) strategies, the procedure first selected a focal essay uniformly from those with the fewest completed comparisons. Random selection then chose uniformly among its eligible partners. For focal essay i and an eligible partner j, the two adaptive strategies evaluated

$$\begin{aligned} \text{p50 strategy:} \quad & \left| p_{ij}^{\mathrm{TS}} - .50 \right|, \\ \text{Pollitt strategy:} \quad & \min\left\{ \left| p_{ij}^{\mathrm{TS}} - \tfrac{1}{3} \right|, \left| p_{ij}^{\mathrm{TS}} - \tfrac{2}{3} \right| \right\}. \end{aligned} \tag{C6}$$

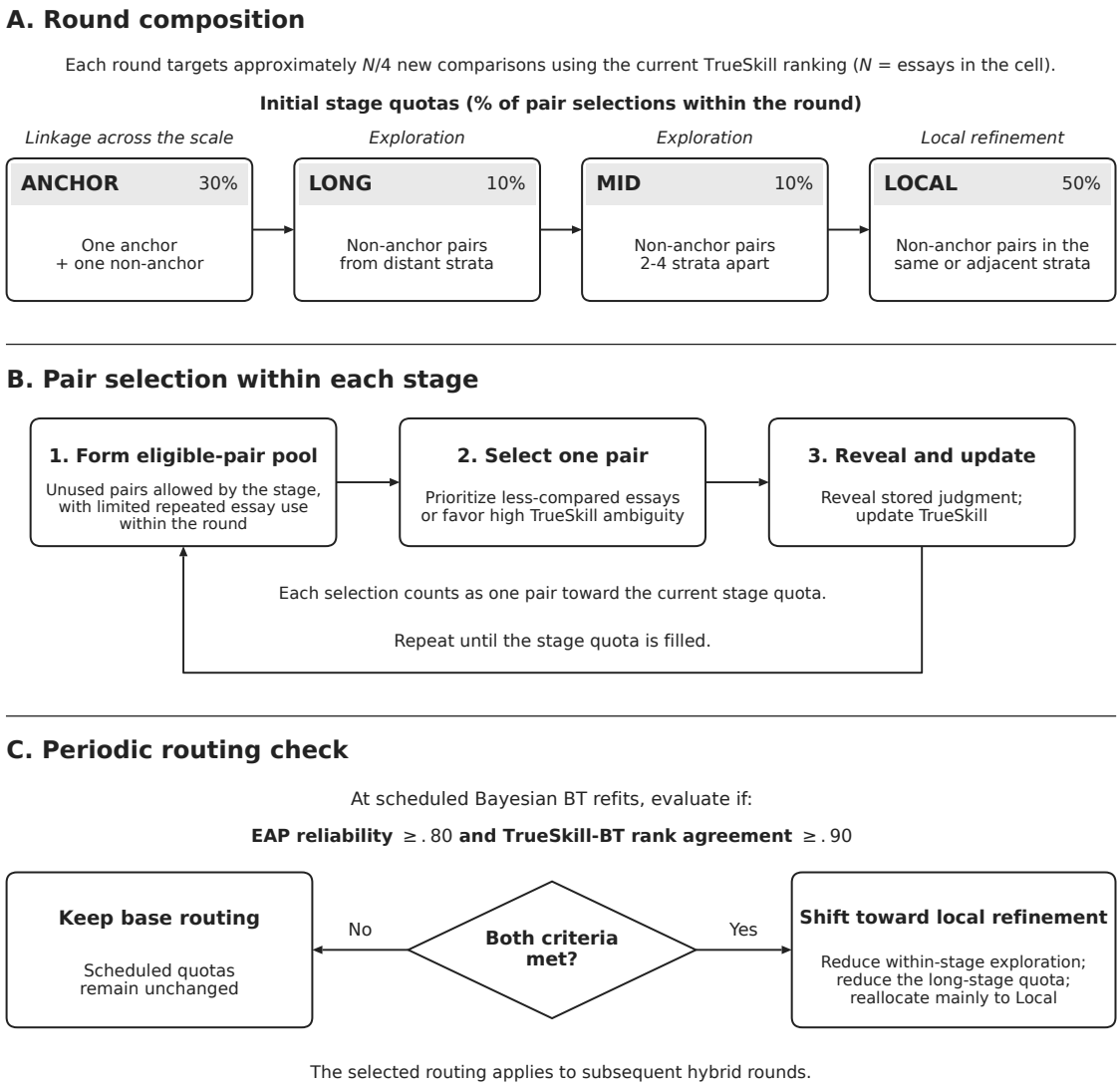
Each strategy selected the eligible partner with the smallest value of its criterion. The p50 strategy therefore targeted the most ambiguous available partner, whereas the Pollitt strategy targeted a less ambiguous comparison. The three strategies differed in partner selection while using the same rule to distribute comparisons across focal essays.

The hybrid strategy used the same evolving TrueSkill information but organized pair selection into rounds, each targeting approximately $N/4$ new comparisons, where N was the number of essays in the cell. Each round proceeded through four stages with complementary purposes: anchor comparisons supported linkage across the quality scale, long and mid comparisons promoted exploration across different regions of the ranking, and local comparisons emphasized refinement among more similar essays. Approximately 10% of essays were designated as anchors, with a minimum of 10 and maximum of 40, distributed approximately 30%, 40%, and 30% across the top, middle, and bottom of the current TrueSkill ranking. For pair routing, the current TrueSkill ranking was divided into 10–17 strata depending on the number of essays in the cell. Anchor comparisons paired one anchor with one non-anchor; long, mid, and local comparisons paired non-anchor essays from widely separated, moderately separated (two to four strata), and same or adjacent rank strata, respectively. During the first four rounds, the anchor, long, mid, and local stages received approximately 30%, 10%, 10%, and 50% of comparisons, respectively; subsequent rounds used 20%, 5%, 10%, and 65%. The overall procedure is illustrated in Supplementary Figure 1.

Within each stage, only previously unused comparisons from the selectable pool were eligible, and the procedure generally limited repeated use of the same essay within a round. Exploration prioritized essays that had received relatively few comparisons, whereas exploitation generally favored pairs with greater TrueSkill ambiguity utility $U_{0,ij}$. If a stage could not supply an eligible pair, predefined fallback rules progressively broadened the candidate search. Periodic Bayesian BT refits informed later routing. Once EAP reliability reached $R_{\mathrm{EAP}} \geq .80$ and the Spearman correlation between the TrueSkill and BT score rankings reached $\rho_{\mathrm{rank}} \geq .90$, the procedure reduced within-stage exploration and the long-stage quota, reallocating most of the reduced quota toward local refinement.

Supplementary Figure 1

Hybrid Pair Selection Rounds, Comparison Stages, and Later-Stage Routing



References

Crossley, S. A., Tian, Y., Baffour, P., Franklin, A., Kim, Y., Morris, W., Benner, M., Picou, A., & Boser, U. (2023). The English Language Learner Insight, Proficiency and Skills Evaluation (ELLIPSE) Corpus. *International Journal of Learner Corpus Research*, *9*(2), 248–269. https://doi.org/10.1075/ijlcr.22026.cro

Herbrich, R., Minka, T., & Graepel, T. (2007). TrueSkill: A bayesian skill rating system. *Advances in Neural Information Processing Systems 19*, 569–576. https://doi.org/10.7551/mitpress/7503.003.0076